

SunoBolo — Voice-Based Inventory Management Using Conversational Speech

Milestone 1.2 · Functional Requirements

Deliverable: Functional Requirements Document

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Prepared by: _____

Project Context

SunoBolo is a voice-first inventory system for small Indian retail shops.

Every voice inventory product on the market works as a **command interface**: you press a button and issue a structured instruction — *"add twenty kilo rice."* That is still data entry, performed with your mouth instead of your thumbs. SunoBolo inverts this. The phone sits on the shop counter and **listens to the shop**, extracting stock movements from the ordinary conversation between owner and customer — sentences that were never addressed to the app at all. A kirana shop already says every transaction out loud, usually twice: once when the customer asks and once when the owner confirms. That speech is a complete, free transaction log that currently evaporates into the air. **SunoBolo captures it.**

Because passive capture is less accurate than dictated commands, nothing it extracts is written to stock automatically. Every extraction becomes a **proposal** the owner approves in bulk, and a conventional **Command Mode** mic button remains available as the reliable path. The novel capability is genuinely novel; the failure mode is an ordinary voice inventory app.

Milestone 1 deliverables

- **1.1** — Requirements Gathering & Analysis
- **1.2** — Functional Requirements (*this document*)
- **1.3** — Technical Design & Architecture
- **1.4** — UI/UX Design & Prototype

Functional Requirements

Priority: **P0** must ship · **P1** if time allows · **P2** documented, deferred

FR-1 — Counter Mode: Passive Voice Capture `P0`

FR-1.1 A single toggle starts Counter Mode. While active, the app listens continuously

and shows a persistent visual indicator that it is listening. No further interaction is

required for the rest of the session.

FR-1.2 Speech is transcribed in rolling segments. Each completed segment is parsed

independently for stock movements. Segments containing no recognisable item are discarded

silently — the app does not clutter the tray with noise.

FR-1.3 One segment may yield **multiple** movements. *"do kilo chawal aur ek Maggi dena"*

produces two proposals from one utterance.

FR-1.4 Nothing extracted in Counter Mode is written to stock. Every extraction becomes a

proposal in the pending tray, carrying its confidence score and the sentence it came from.

FR-1.5 Counter Mode restarts recognition automatically when the browser ends a session on

silence, so that listening survives quiet periods without user action.

Acceptance: With Counter Mode on and no buttons pressed, speaking *"bhaiya do kilo chawal*

dena" places a proposal reading Rice, 2, kg, OUT in the tray within 3 seconds.

FR-2 — Command Mode: Deliberate Entry `P0`

FR-2.1 A mic button accepts a single, deliberate utterance for a well-formed entry —

used for goods received, for corrections, and as the reliable path when the counter is noisy.

FR-2.2 Command Mode understands four intents:

Intent	Meaning	Example
STOCK_IN	Goods received	<i>"bees kilo chawal aaya"</i>
STOCK_OUT	Goods sold	<i>"do dozen anda becha"</i>
QUERY_ITEM	Stock of one item	<i>"chawal kitna hai"</i>
QUERY_LOW	What needs reordering	<i>"kya khatam ho raha hai"</i>

FR-2.3 Command Mode results are shown on a confirmation card and committed on one tap.

Because the utterance is deliberate and well-formed, accuracy here is materially higher than in Counter Mode, which is why it remains the reliable floor beneath the passive feature.

Acceptance: *"bees kilo chawal aaya"* produces a confirmation card reading Rice, 20, kg, IN within 3 seconds of speech ending.

FR-3 — Regional and Mixed-Language Speech `P0`

FR-3.1 Hindi (**hi-IN**), Telugu (**te-IN**), Indian English (**en-IN**), chosen by a persistent three-way picker that sets both the recognition locale and the language of all replies.

FR-3.2 Within one utterance the item, the unit and the verb may each independently be in a different language. Matching is per-token against a multilingual alias list; the sentence is never required to be monolingual.

FR-3.3 Trade units understood in all three languages, singular and plural:

Base	Accepted trade units
kg	kilo, kilogram, किलो, కీలో, gram, quintal, क्विंटल, bag, bori, बोरी, గోసు
pc	piece, nag, నగ, dozen, दर्जन, డజను, carton, peti, పేటీ, box, dabba, డబ్బా, tray
litre	litre, liter, లీటర్, లీటరు, ml, tin

FR-3.4 Number words — *ek/do/teen/paanch/bees/sau, okati/rendu/moodu/aidu/nooru* —

including compounds (*"teen sau"* → 300) and fractional trade words *dhai* 2.5, *derh* 1.5, *sawa* 1.25, *paune* 0.75.

FR-3.5 Replies are rendered and spoken in the selected language using plain trade vocabulary. No software terms (*record, entry, transaction, SKU, sync*) appear anywhere.

Acceptance: *"rendu bori biyyam vacchindi"* registers +100 kg of Rice and the app replies in Telugu with the new total.

FR-4 — Pending Tray, Correction and Search `P0`

FR-4.1 All Counter Mode proposals collect in a tray, newest first, each showing the item, quantity, unit, direction, confidence, and the sentence it was extracted from.

FR-4.2 Bulk approval. A single control accepts every proposal above the confidence threshold. Reviewing twenty proposals takes seconds; typing twenty entries does not.

FR-4.3 Confidence gate.

Item match score	Behaviour
≥ 85	Proposal shown ready to accept; included in bulk-accept
60–85	Shown with the next two candidate items as one-tap alternatives; excluded from bulk-accept
< 60	Discarded silently — the system does not guess in passive mode
Top two within 5 points	Forced "did you mean" regardless of absolute score

FR-4.4 Tap correction. Quantity, unit, item and direction are each individually editable.

Quantity uses steppers, unit and direction use chips, item uses the search list. No keyboard.

FR-4.5 Item search filters the catalogue by any alias in any language, so "rice",

"chawal" and "ಬಿయ್ಯం" all find the same item.

FR-4.6 Manual entry provides a typed path to the same confirmation card, for noisy environments and unsupported browsers, writing through the identical parse and commit path.

Acceptance: A misheard proposal is corrected to the right item and accepted using taps only, with the on-screen keyboard never appearing.

FR-5 — Dashboard, Alerts and Reorder Suggestions `P0`

FR-5.1 The home screen lists every item with its current quantity in the item's display

unit, sorted so items needing attention appear first, each with a status colour: green

healthy, amber low, red out.

FR-5.2 An item is *low* when `current_qty <= reorder_level`. Low items are counted in a home-screen banner and listed in full on the Alerts screen.

FR-5.3 Each low item carries a reorder suggestion restoring stock to a target level, expressed in the item's natural **purchase** unit — "order 2 bori", not "order 100 kg".

FR-5.4 Item detail shows current stock, pack sizes, reorder level, last price, and recent transaction history — each entry showing what was said, when, and the resulting change.

Acceptance: Rice at 40 kg with reorder level 50 appears amber, is counted in the banner, and suggests "order 2 bori (100 kg)".

FR-6 — Clear Confirmations and Summaries `P0`

FR-6.1 After a commit the app states what changed and the new total, on screen and aloud —

"20 किलो चावल जोड़ा गया। अब स्टॉक 145 किलो।"

FR-6.2 Stock questions are answered in one spoken sentence and shown on screen simultaneously.

FR-6.3 Undo. The most recent commit reverses with one tap. The reversal is recorded as its own transaction; nothing is deleted.

FR-6.4 All strings come from a per-language template table. Numbers render as digits, which all three user groups read reliably.

FR-7 — Privacy, Persistence and Reliability `P0`

FR-7.1 No audio retention. Speech recognition runs entirely in the browser. ****No audio is recorded, stored, or transmitted.**** Only extracted values and the resulting text transcript leave the device.

FR-7.2 Visible listening state. Whenever Counter Mode is active, an unmistakable indicator is shown. The app never listens without saying so.

FR-7.3 Shop login. Shop name plus a 4-digit PIN, stored hashed. Session remembered on the device.

FR-7.4 Server-side persistence. Items and transactions live in a managed database, not browser storage. Data survives reload, device change and server restart.

FR-7.5 Immutable ledger. Stock is never overwritten. Every change is an append-only row; current stock is derived by summing them. Corrections are new rows, so every number is traceable to the sentence that caused it.

FR-7.6 Backup. Catalogue and full history exportable as CSV.

Edge Cases and Defined Behaviour

Case	Behaviour
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Segment contains no known item	Discarded silently, not shown in the tray
Item not in catalogue	Offer "add new item" with the heard name pre-filled
Quantity missing	Default to 1, highlight the field for review
Unit missing	Use the item's base unit and show it
Direction ambiguous at the counter	Default to OUT — counter conversation is overwhelmingly selling — and show the IN/OUT chips
Stock-out would go negative	Allow but flag; physical stock and records genuinely diverge
Two items within 5 points	Force "did you mean" regardless of score
Same sentence recognised twice	Deduplicate identical extractions inside a short window
Browser has no speech support	Open manual entry automatically with an explanatory line

Traceability to the Brief

Brief requirement	Covered by
1 — Voice input for add/remove/update/query	FR-1, FR-2
2 — Regional & mixed language, trade units	FR-3
3 — Dashboard, alerts, reorder suggestions	FR-5
4 — Minimal typing, correction, search	FR-4
5 — Clear confirmations and summaries	FR-6
Tech 1 — Multilingual speech-to-text	FR-1.1, FR-3.1
Tech 2 — Data model, unit conversion, history	FR-3.3, FR-7.5
Tech 3 — Mobile-first lightweight web app	FR-4, FR-5
Tech 4 — Backend APIs for tracking, updates, alerts	FR-5, FR-7.4
Tech 5 — Auth, backup, logging	FR-7